

Birch and Swinnerton-Dyer Under the Microscope: Exact Rank and Sha Checks for 1,328 Elliptic Curves, and What Learning Sees

PARI/GP computation with a PINNeAPLe learning probe — evidence, not proof

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Abstract

The Birch and Swinnerton-Dyer (BSD) conjecture relates the rank of the Mordell–Weil group $E(\mathbb{Q})$ of an elliptic curve to the order of vanishing of its L-function at $s = 1$, and predicts the leading Taylor coefficient in terms of arithmetic invariants including the order of the Tate–Shafarevich group III . We take all non-singular curves $y^2 = x^3 + ax + b$ with $|a|, |b| \leq 25$, remove isomorphic duplicates (1,328 curves; conductors 17–1,248,832), and compute with PARI/GP: the analytic rank, the algebraic rank by 2-descent, and — for ranks 0 and 1 — the analytic order of III predicted by the BSD formula. The algebraic rank is determined in all 1,328 cases (distribution 499/692/134/3 for ranks 0/1/2/3) and equals the analytic rank in 1,328/1,328. In all 1,191 rank-0/1 cases the predicted III is a perfect square to within $4.4\text{e-}16$ (values 1, 4 and 9). A validation-selected PINNeAPLe MLP predicts the rank from 95 normalised Frobenius traces with 80.4% accuracy versus 78.9% for a Mestre–Nagao sum. For analytic rank ≤ 1 the rank equality is already a theorem (Gross–Zagier, Kolyvagin, modularity); these computations are consistency evidence, not new mathematics.

Keywords: Birch and Swinnerton-Dyer; elliptic curves; L-functions; Tate–Shafarevich group; PARI/GP; machine learning; Millennium Prize problems

What this study shows — and what it does not

- Shows: for 1,328 curves, analytic and algebraic ranks agree; the BSD-predicted III is always a perfect square (as it must be when III is finite).
- Does not show BSD: ranks ≤ 1 are covered by theorems already, and for rank ≥ 2 the analytic rank is a numerical determination of the order of vanishing (a certified proof of $L(E,1) = L'(E,1) = 0$ is a different matter).
- The learning result shows that Frobenius traces carry rank information (known: Mestre–Nagao, murmurations), not that rank is computable this way.

1 Background and strategy

For $E/\mathbb{Q}$ with L-function $L(E,s)$, BSD states $\text{ord}_{s=1} L(E,s) = \text{rank } E(\mathbb{Q}) = r$ and $L^{(r)}(E,1)/r! = \Omega_E \cdot \text{Reg}_E \cdot |\text{III}| \cdot \prod_p c_p / |E(\mathbb{Q})_{\text{tors}}|^2$. Numerically we can (i) compare the numerically determined order of vanishing with a rank certified by descent, and (ii) solve the formula for III and test whether it is a positive integer square, a strong non-trivial consistency condition.

2 Methods

- Curves: all (a, b) with $|a|, |b| \leq 25$ and $4a^3 + 27b^2 \neq 0$, reduced to minimal models; duplicates removed by (j-invariant, discriminant).
- Analytic rank and $L^{(r)}(E,1)$: *ellanalyticrank*. Algebraic rank: *ellrank* (2-descent bounds; 'proven' when lower = upper bound).
- Regulator: canonical-height matrix of generators after *ellsaturation* (so the generators span $E(\mathbb{Q})/\text{tors}$, not a finite-index subgroup). $\Omega \cdot \prod_p c_p / |\text{tors}|^2$: *ellbsd*.
- Learning: features $a_p/\sqrt{p}$ for the 95 primes $p < 500$; target rank (0, 1, 2); 70/15/15 split; PINNeAPLe registry candidates selected on validation; baseline: Mestre–Nagao sum $\sum_p a_p \log p / p$ with class means fitted on training curves.

3 Results

Table 1. BSD checks (PARI/GP).

Quantity	Value
curves (distinct)	1,328
rank certified by descent	1,328
rank distribution 0/1/2/3	499 / 692 / 134 / 3
analytic rank = algebraic rank	1,328 of 1,328
rank-0/1 curves with $ \text{III} $ checked	1,191
max distance of $ \text{III} _{\text{an}}$ to a square	4.4e-16
$ \text{III} _{\text{an}}$ values (count)	1: 1159, 4: 27, 9: 5
CPU time	6.8 h

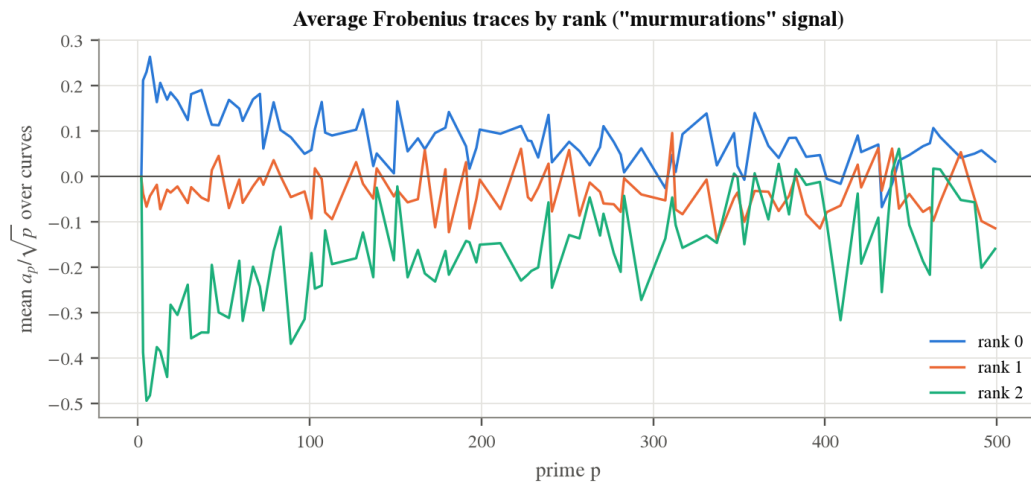


Figure 1. Average of $a_p / \sqrt{p}$ over curves of each rank. Higher rank goes with systematically negative traces at small p — the classical Mestre–Nagao heuristic, recently rediscovered as 'murmurations' by machine-learning studies.

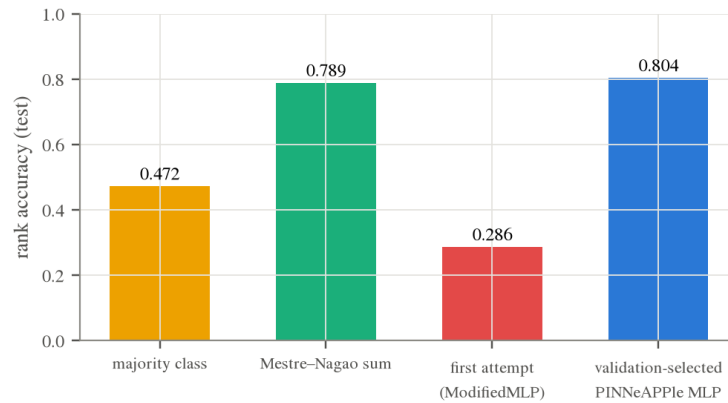


Figure 2. Rank prediction accuracy on held-out curves.

The selected model (bench_mlp(w256,d4,geLU)) confusion matrix (rows = true rank 0/1/2): $\begin{bmatrix} 65 & 22 & 0 \\ 13 & 79 & 2 \\ 0 & 2 & 16 \end{bmatrix}$. The first attempt, a single Fourier-feature ModifiedMLP with a fixed bandwidth, scored 28.6% — below the majority class — and is reported for transparency; it was replaced by validation-based model selection, not by looking at test scores.

4 Discussion

Every computed number agrees with BSD, as expected from the theorems that cover ranks 0 and 1 and from decades of computations (e.g. the Cremona and LMFDB databases). Where computation adds something is scale and automation; where it cannot help is the proof, which needs to control III and the L-function in general. The learning probe reproduces a known heuristic signal with slightly better accuracy than the hand-crafted sum; it offers no mechanism.

References

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